



Name of the module: DFT
Location in GIT repository: vspa_common\vspa-lib\dft

Overview:

This module provides a kernel for a Generic DFT. The generic DFT has the following expression:

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}}$$

where:

$x(n)$ – input signal
 N – input sequence length

Assembly Functions API:

```
void dft_hfx_hfl_asm ( cfixed16_t *in_p,  
                      cfixed16_t *scratch_p,  
                      cfloat16_t *out_p,  
                      uint32_t    n_dft);  
  
void dft_sfl_hfl_asm( cfixed16_t *in_p,  
                      cfixed16_t *scratch_p,  
                      cfloat16_t *out_p,  
                      uint32_t    n_dft);  
  
void dft_hfx_sfl_asm( cfixed16_t *in_p,  
                      cfloat16_t *out_p,  
                      uint32_t    n_dft);  
  
void dft_sfl_sfl_asm( cfixed16_t *in_p,  
                      cfloat16_t *out_p,  
                      uint32_t    n_dft);  
  
void mini_dft_hfx_hfl_asm( cfixed16_t *in_p,  
                          cfloat16_t *out_p,  
                          uint32_t    n_dft);  
  
void mini_dft_sfl_hfl_asm( cfloat32_t *in_p,  
                          cfloat16_t *out_p,  
                          uint32_t    n_dft);  
  
void mini_dft_hfx_sfl_asm( cfixed16_t *in_p,  
                          cfloat32_t *out_p,  
                          uint32_t    n_dft);  
  
void mini_dft_sfl_sfl_asm( cfloat32_t *in_p,  
                          cfloat32_t *out_p,  
                          uint32_t    n_dft);
```



in_p - Pointer of the DFT input
scratch_p - Pointer of the Scratch
out_p - Pointer of the DFT output
n_dft - Input sequence length

DMEM requirements

- For dft_xx_xx_asm (Support sequence length >96)

Variable	Minimum size (half-words)	Minimum alignment (half-words)	Allocated by
in_p	194/388*	64	Caller
scratch_p	388	64	Caller
out_p	194/388*	64	Caller

* 194 for half_float/half_fixed precision and 388 for single precision.

- For mini_dft_xx_xx_asm (Support sequence length 2:96)

Variable	Minimum size (half-words)	Minimum alignment (half-words)	Allocated by
in_p	2/4*	64	Caller
out_p	2/4*	64	Caller

* 2 for half_float/half_fixed precision and 4 for single precision.

Test plan:

- Test random input signal in time domain (size = 98 - 839).